



P R O J E C T R E P O R T

WildWatch PK

Pakistan Wildlife Extinction Risk Analyzer

Machine Learning for Wildlife Conservation in Pakistan

Project Title	WildWatch PK
Category	Machine Learning + Data Science
Domain	Wildlife Conservation
Algorithm Used	Random Forest Classifier
Dataset	42 Species — Custom Dataset
Tools	Python, Scikit-Learn, Streamlit, Plotly
Developed by	Malayika Tahir
Supervised by	Mr. Abdul Aziz
Report Type	Academic Project Report

Protecting Pakistan's Biodiversity Through Artificial Intelligence

SECTION 01

Introduction

Pakistan is a country of extraordinary natural diversity. From the snow-capped peaks of the Karakoram and Himalayas to the vast plains of Punjab, the mangrove forests of Sindh, and the arid deserts of Balochistan, Pakistan is home to thousands of plant and animal species. Many of these species are found nowhere else on Earth.

However, Pakistan's wildlife faces serious and growing threats. Rapid urbanization, deforestation, climate change, illegal hunting, habitat destruction, and pollution have placed many species in grave danger. The Snow Leopard, Markhor, Indus River Dolphin, Marco Polo Sheep, and several species of migratory birds are just a few examples of animals that may be approaching extinction if conservation efforts are not strengthened.

Despite these challenges, wildlife monitoring in Pakistan largely depends on manual surveys, fragmented reports, and limited institutional capacity. There is a significant gap between the data that exists about wildlife and the tools available to analyze and act on that data quickly and accurately.

WildWatch PK was created to fill this gap, to bring the power of machine learning into wildlife conservation and help identify which species are most at risk before it is too late.

This project demonstrates how data science can be applied to environmental challenges. By building a machine learning model trained on species characteristics and population data, WildWatch PK can classify animals into different extinction risk categories and provide meaningful insights for conservationists, researchers, and government departments.

SECTION 02

Problem Statement

Wildlife conservation in Pakistan faces a fundamental data problem: we know that species are declining, but we lack fast, scalable, and data-driven tools to identify which species need urgent help, how severe the risk is, and where conservation resources should be directed.

Core Problems Identified

- No centralized, real-time wildlife risk monitoring system exists in Pakistan
- Manual field surveys are slow, expensive, and often incomplete
- Conservation decisions are reactive rather than predictive

- Species populations are tracked inconsistently across different regions
- Limited use of modern data science and machine learning in environmental departments
- Policymakers lack actionable, visual intelligence reports on threatened species

Impact of the Problem

The consequences of this gap are severe. Species that could have been saved with early intervention are lost. Conservation funding is misallocated. Government departments cannot prioritize effectively. The rich biodiversity that defines Pakistan's natural heritage continues to erode.

The core question this project addresses: Can machine learning accurately classify the extinction risk of animal species using biological and population data, and can this intelligence be delivered in a form useful to forest officers and policymakers?

SECTION 03

Why We Chose This Project

Emotional Motivation

There is something deeply moving about the idea that an animal species, one that has existed for thousands or millions of years, can disappear forever within our lifetime because of human activity. The Snow Leopard, Pakistan's national animal, exists in fewer than 200 individuals in our mountains. The Indus River Dolphin is down to a few hundred. These are not abstract statistics. These are living creatures facing extinction while we have the knowledge and tools to help.

This project was chosen because it connects technology with something that truly matters. It is not a theoretical exercise. Every prediction the model makes could represent a real species, a real ecosystem, and a real opportunity to intervene before it is too late. The emotional weight of that possibility made this project feel worth the effort.

"If we can teach a machine to recognize cancer in a medical scan, we can teach it to recognize extinction risk in an ecosystem. The urgency is the same."

Logical Motivation

From a data science perspective, wildlife conservation is an ideal machine learning problem. The data is structured, the features are meaningful, the classification task is clear, and the real-world application is direct. There are several logical reasons why this project was selected:

Reason	Explanation
Real Impact	The model can assist actual conservation decisions, not just academic exercises
Rich Dataset	Biological data (lifespan, population, speed, weight) provides meaningful ML features
Clear Problem	Multi-class classification with 3 risk levels is well-defined and measurable
Interdisciplinary	Combines biology, data science, and public policy into one coherent system
Scalable Concept	The approach can be extended to hundreds of species across South Asia
Visual Output	Results can be communicated through dashboards accessible to non-technical users

Choosing a project at the intersection of machine learning and environmental science also reflects a belief that data scientists have a responsibility to apply their skills to problems beyond commercial applications. Wildlife conservation is one of the most critical challenges of our time, and this project is a small but meaningful contribution to that effort.

SECTION 04

Project Overview

WildWatch PK is a machine learning-based wildlife conservation system designed to analyze species characteristics and predict their extinction risk level. The project takes raw wildlife data as input, processes it through a complete machine learning pipeline, and delivers predictions and visualizations that can be used by forest officers, researchers, and conservation organizations.

Project Name & Meaning

WildWatch: Refers to the active monitoring and surveillance of wildlife, keeping a watch on species that may be in danger.

PK: Pakistan, clearly indicating the geographic focus and national relevance of the project.

Core Objectives

- Build a machine learning model that classifies animal species into extinction risk categories
- Analyze wildlife data to discover patterns between species characteristics and risk level
- Create an interactive dashboard for forest officers and policymakers to explore results
- Identify which species in Pakistan require urgent conservation attention
- Demonstrate that data science can be applied meaningfully to environmental problems

System Output

The system produces three types of output for each animal species:

- Risk Classification: Low Risk (0), Medium Risk (1), or High Risk (2)
- Population Analysis: Visual comparison of current population vs historical range
- Officer Reports: Field-ready briefings for forest department use

SECTION 05

Dataset Description

A custom dataset was created for this project containing information about animal species relevant to Pakistan's biodiversity. The dataset was assembled from publicly available biodiversity databases, wildlife surveys, and conservation organization reports.

Dataset Summary

Total Species	42 Animal Species
Total Features	25 Columns
Target Variable	Extinction_Risk (0 = Low, 1 = Medium, 2 = High)
Geographic Focus	Pakistan and South Asia
Data Type	Structured / Tabular
Population Range	200 (Snow Leopard) to 3,000,000 (Grey Wolf)

Key Features Used in Model Training

Feature Name	Data Type	Description
Min-Population size	Integer	Minimum known historical population
Max-Population size	Integer	Maximum known historical population
Min-Life Span	Integer	Minimum lifespan in years
Max-Life Span	Integer	Maximum lifespan in years
Min-Length (M)	Float	Minimum body length in meters
Min-Wingspan (cm)	Float	Minimum wingspan in centimeters
Current_population	Integer	Current estimated population (KEY)

Target Variable, Extinction Risk

The target variable was engineered from the Current_population field using the following thresholds:

Risk Level	Population Threshold	Label
0 — Low Risk	Above 100,000	Stable
1 — Medium Risk	10,000 to 100,000	Under Stress
2 — High Risk	Below 10,000	Critical Danger

SECTION 06

Machine Learning Pipeline

The project follows a complete, end-to-end machine learning pipeline. Each step was carefully executed to ensure the model produces reliable and meaningful predictions.

Step 1 : Data Collection

Wildlife information was gathered from publicly available biodiversity databases and conservation reports. A custom dataset was constructed covering 42 animal species with 25 attributes each, including biological measurements, population estimates, and taxonomic classifications.

Step 2 : Data Preprocessing

Raw data is rarely clean or ready for machine learning. Several preprocessing steps were applied:

- Missing values in categorical columns were filled using the most frequent value (mode)
- Missing values in numerical columns were filled using the column average (mean)
- Duplicate records were identified and checked
- Unnecessary columns such as Animal ID, Kingdom, Phylum, Subphylum, and Class were dropped as they did not contribute predictive value
- Population size columns containing comma-separated text values were cleaned and converted to integers

Step 3 : Feature Engineering

A critical step in the pipeline was the creation of the target variable. Since the dataset did not include a pre-labeled extinction risk column, one was engineered using the Current_population field:

```
df['Extinction_Risk'] = 0 # Default: Low Risk
df.loc[df['Current_population'] < 10000, 'Extinction_Risk'] = 2
df.loc[(pop >= 10000) & (pop < 100000), 'Extinction_Risk'] = 1
```

Step 4 : Encoding & Scaling

Machine learning models require numerical inputs. All categorical text columns (such as species Order, Family, Genus) were encoded using Label Encoding. Feature values were then normalized using MinMaxScaler to bring all features to the same scale (0 to 1), preventing any single feature from dominating the model.

```
scaler = MinMaxScaler() X_scaled = scaler.fit_transform(X)
```

Step 5 : Train/Test Split

The dataset was split into training data (80%) and testing data (20%) using stratified splitting to ensure all three risk classes were proportionally represented in both sets. The random_state=42 parameter ensures reproducibility.

Step 6 : Exploratory Data Analysis (EDA)

Before training the model, exploratory analysis was conducted to understand the data. The following visualizations were used:

- Heatmap: To visualize correlations between numerical features
- Boxplots: To analyze the distribution of population values across risk levels
- Distribution plots: To understand how features like lifespan and weight are spread
- Bar charts: To show species count by class and order
- Scatter plots: To identify relationships between population and risk

Step 7 : Model Training

The core machine learning algorithm selected for this project is the Random Forest Classifier. This decision was made for several reasons explained in the next section. The model was trained on the scaled feature matrix with the engineered Extinction_Risk target.

Step 8 : Prediction & Deployment

After training, the model was integrated into an interactive Streamlit dashboard called WildWatch PK. The dashboard allows forest officers and researchers to explore predictions, view species details, and access field briefing reports, all powered by the trained Random Forest model.

SECTION 07

Model Selection (Random Forest)

Among all machine learning classification algorithms considered, the Random Forest Classifier was selected as the primary model for WildWatch PK. This choice was not arbitrary, it was based on the specific characteristics of the dataset and the nature of the problem.

What is a Random Forest?

A Random Forest is an ensemble learning method that constructs many decision trees during training and combines their outputs to produce a final prediction. Instead of relying on a single tree (which is prone to overfitting), the algorithm builds a 'forest' of trees, each trained on a random subset of the data and features. The final class is decided by majority vote.

Why Random Forest Was Chosen

Reason	Detail
Handles Small Data	Works well with limited samples (42 species) without overfitting severely
Feature Importance	Identifies which biological features most influence extinction risk
Multi-class Support	Natively handles 3-class classification (Low / Medium / High)
Robust to Outliers	Population data has extreme outliers; Random Forest handles them well
No Feature Assumptions	Does not assume normal distributions or linear relationships
Class Imbalance	class_weight='balanced' handles unequal class distribution
Interpretable Output	Provides probability scores and feature importance rankings

Model Configuration

```
RandomForestClassifier(      n_estimators = 100,      # 100 decision trees
random_state = 42,          # Reproducibility      class_weight = 'balanced'
# Handle imbalanced classes )
```

Feature Importance Results

After training, the model revealed which features contributed most to predicting extinction risk. The most important feature was Current_population, which makes intuitive sense, a species with a very low population is directly at risk. Population size range features (Min and Max historical population) were also highly informative.

SECTION 08

Tools & Technologies Used

Tool / Library	Version	Purpose
Python	3.10+	Core programming language for all development
Pandas	2.x	Data loading, cleaning, transformation, and analysis
NumPy	1.x	Numerical operations and array manipulation
Scikit-Learn	1.x	Machine learning, Random Forest, MinMaxScaler, metrics
Matplotlib	3.x	Static charts and EDA visualizations in notebook
Seaborn	0.x	Statistical plots , heatmaps, boxplots, distributions
Streamlit	1.x	Interactive web dashboard and frontend interface
Plotly	5.x	Interactive charts , donut, bar, scatter, gauge in dashboard
Jupyter	Notebook	Exploratory data analysis and model development

Development Environment

- Code Editor: PyCharm / Jupyter Notebook
- Operating System: Windows 10/11
- Package Manager: pip / conda
- Version Control: Git

SECTION 09

WildWatch PK (Streamlit Dashboard)

The trained model was integrated into a professional, interactive web dashboard built with Streamlit. The dashboard is designed to serve two user types: data scientists who want to explore the model, and field-level forest officers who need actionable intelligence.

Dashboard Pages

Page	Content
Mission Control	Overview dashboard — KPI cards, donut chart, population bar chart, species class risk
Threat Assessment	Tabbed view of all species by risk level with photos and population data
Forest Officer Briefing	Field intelligence report — priority actions, bilingual (Urdu/English), photo cards
ML Model Analysis	Feature importance, confusion matrix, precision/recall/F1 metrics
Species Lookup	Search any species — view photo, biological data, risk gauge visualization

Design Philosophy

- Futuristic 2030-style dark theme with deep forest green color palette
- Orbitron monospace font for technical metrics, Plus Jakarta Sans for body text
- Animal photos sourced from Wikipedia/Wikimedia for each species
- Color-coded risk system: Red (High), Amber (Medium), Green (Stable)
- Mobile-responsive layout with interactive Plotly charts



SECTION 10

Results & Model Performance

The Random Forest model was evaluated on the held-out test set (20% of data). Performance was measured using accuracy score, confusion matrix, and a full classification report covering precision, recall, and F1-score for each of the three risk classes.

Model Accuracy

The trained Random Forest Classifier achieved strong classification performance on the test set, demonstrating that biological and population features are meaningful predictors of extinction risk.

What the Results Mean

- High precision for High Risk class: When the model says a species is critically endangered, it is usually correct
- Strong recall for High Risk: The model catches most truly endangered species ,critical for conservation
- The confusion matrix shows most errors are between adjacent risk levels (Low/Medium or Medium/High) rather than extreme misclassifications
- Class weighting (balanced) ensured the model did not ignore the minority High Risk class

Predicted Risk Distribution

Risk Category	Count	Percentage
High Risk (2)	~6 species	~14%
Medium Risk (1)	~12 species	~29%
Low Risk (0)	~24 species	~57%

SECTION 11

Key Species Findings

Based on the model's predictions and population data analysis, the following species were identified as requiring the most urgent conservation attention in Pakistan:

Critical — High Risk Species

Species	Population	Primary Threat	Habitat
Snow Leopard	~200	Poaching, habitat loss	Karakoram, Himalaya
Bengal Tiger	~5,300 global	Deforestation, conflict	Mangrove forests
Markhor	~6,000	Trophy hunting, competition	Chitral, GB mountains
Bearded Vulture	~2,500 global	Lead poisoning, habitat	Northern mountains

Medium Risk — Under Monitoring

- Blue Whale (~25,000) — Ship strikes in Arabian Sea
- Mugger Crocodile (~10,000) — Indus River habitat degradation
- Eurasian Lynx (~50,000) — Forest fragmentation

Stable Species — Routine Monitoring

- Grey Wolf (~3,000,000) — Currently stable but prey base declining
- Golden Eagle (~250,000) — Protected cliff nesting zones needed
- Brown Bear (~200,000) — Seasonal corridor management required

SECTION 12

Project Summary

WildWatch PK began as an idea: what if machine learning could help Pakistan protect its wildlife more effectively? What started as a data science exercise became something with genuine real-world meaning. By the end of development, the project had evolved into a complete AI-powered conservation intelligence system.

Project Name	WildWatch PK, Pakistan Wildlife Extinction Risk Analyzer
Core Problem	Lack of data-driven extinction risk monitoring in Pakistan
Solution	ML pipeline + interactive Streamlit dashboard
Algorithm	Random Forest Classifier (100 trees, balanced class weights)
Dataset	42 species, 25 features, custom-built dataset
Key Features	Current population, population range, lifespan, length, wingspan
Target Variable	Extinction_Risk (0=Low, 1=Medium, 2=High)
Dashboard Pages	5 interactive pages (Mission Control to Species Lookup)
Key Finding	6 High Risk species identified requiring immediate intervention
Most At Risk	Snow Leopard (~200 individuals), Markhor (~6,000)

SECTION 13

Conclusion

WildWatch PK is more than a machine learning exercise. It is an attempt to prove that data science can serve a purpose beyond commercial optimization, that it can be directed toward protecting the natural world that we share with thousands of other species.

Through this project, we demonstrated that a well-designed machine learning pipeline, even trained on a relatively small dataset of 42 species, can produce meaningful extinction risk classifications with real conservation value. The Random Forest algorithm proved to be the right tool for this problem, robust, interpretable, and effective on structured biological data.

The Streamlit dashboard transforms raw predictions into accessible intelligence. Forest officers who may never use a Jupyter notebook can now open a browser, select a species, and immediately understand its risk level, population status, and recommended action, in both English and Urdu.

Pakistan's Snow Leopard, Markhor, and dozens of other species are running out of time. WildWatch PK is one small step toward making sure we do not lose them without a fight, and that the fight is informed by data.

Future Scope

- Expand dataset to 200+ species across all of South Asia
- Integrate satellite data and climate change indicators as features
- Develop a mobile application for field rangers
- Connect with government wildlife databases for real-time updates
- Add species habitat mapping using GIS data
- Deploy as a public API for conservation organizations

WildWatch PK (Protecting Pakistan's Biodiversity Through Artificial Intelligence)

Machine Learning for Wildlife Conservation | Academic Project Report

By
Malayika Tahir